

A photograph of a custom electronic circuit assembled on a white breadboard. The central component is a Raspberry Pi Zero microcontroller board. To its left, a small blue USB-to-UART bridge module is connected. Above the Pi Zero, a silver USB microphone is plugged into a USB port. Two red LEDs are visible: one on the left, connected to a blue USB-to-UART module, and another on the right, connected to the Pi Zero's GPIO pins. Various colored wires (red, white, purple, blue, yellow, green) are used to connect the components to the breadboard's power rails. The entire setup is placed on a dark wooden surface.

The screenshot shows the Arduino IDE interface with the following components:

- Top Bar:** File, Edit, Sketch, Tools, Help. The project name is "DOIT ESP32 DEVKIT V1".
- Code Editor:** Displays the code for "1_air_quality_monitor.ino". The code defines Blynk credentials, pins, and DHT sensor settings.
- Serial Monitor:** Shows the output of the code, displaying "Air Quality: 2017" and temperature readings.

```
1 #define BLYNK_TEMPLATE_ID "TMPL35oZyn_g0"
2 #define BLYNK_TEMPLATE_NAME "Air Quality Monitoring"
3 #define BLYNK_AUTH_TOKEN "buAQDEo4hbl1POT7x4d4Z-AOR3CRKL"// ⚠️ Add this from Blynk
4
5 #include <WiFi.h>
6 #include <BlynkSimpleEsp32.h>
7 #include "DHT.h"
8
9 // WiFi credentials
10 char ssid[] = "project1";
11 char pass[] = "123456789";
12
13 // Pins
14 #define MQ135_PIN 34
15 #define LED_PIN 2
16 #define BUZZER_PIN 15
17
18 #define DHTPIN 4
19 #define DHTTYPE DHT11
20
21 DHT dht(DHTPIN, DHTTYPE);
```

Output Serial Monitor X

Message (Enter to send message to 'DOIT ESP32 DEVKIT V1' on 'COM9')

New Line 115200 baud

```
22:30:39.456 -> *****Air Quality: 2017
22:42:13.075 -> Temperature: 0.40
22:42:45.104 -> Air Quality: 1745
22:42:45.104 -> Temperature: 33.40
```

Ln 10, Col 24 DOIT ESP32 DEVKIT V1 on COM9 10:42 PM 4/18/2020

Output Dashboard

